Appendix 1 exercises

1. Simplify $(-8)^{2/3}, 16^{-3/4}, (-16)^{1/2}$, stating whether each is defined over the reals.
2. From $E(x)E(y) = E(x+y)$, prove $E(-x) = 1/E(x)$ and nonvanishing for every real x.
3. For $a, b > 0, b \neq 1, c > 0$, prove $a^{\log_b c} = c^{\log_b a}$.
4. Give an exact decomposition and approximation of $\log_2 48$. Explain its constant ratio to $\log_{10} 48$.
5. Solve $\ln(x-1) + \ln(x+2) = \ln 4$, stating the original domain first.
6. Prove $e^x \geq 1 + x$ for every real x, with equality conditions. Use it to compare e^{-2} with minus one.
7. Use uniqueness of positive q-th roots to prove rational powers independent of fraction representation. Give a negative-base nested-power counterexample with both sides real and defined.
8. Solve $\log_{1/2}(x-1) \geq 2$, stating the domain and checking one value inside and one outside the solution set.
9. Without L'Hôpital's rule, prove $n^r = o(b^n)$ for fixed $r > 0, b > 1$, and $\ln n = o(n^\varepsilon)$ for $\varepsilon > 0$. Why does allowing $b = 1 + 1/n$ break the first conclusion?
10. Find bit length and ceiling-halving rounds to one for $n = 1, 8, 13, 16$. Express $\ln(e^{1000} + e^{999})$ without overflow, giving intermediate ranges.
11. For $n = 16$, compute $(\log_2 n)^2$, $\log_2(n^2)$, and $\log_2^* n$. Stop iterating when the value is at most 1.

References and attribution

Lin Cing, handwritten data structures notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 1.

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https://github.com/xinhi6716/computer-science-self-study